

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Thunder Client
Type of Testing	API Response / Functional Performance Check
Target Module / API	AI Blog Generation API – POST /api/ai/generate-blog
Test Environment	Local – http://127.0.0.1:5000
Test Date	25 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	AI blog generation API request using topic and category	1	Single request	API returns generated blog content successfully
2	API response verification	1	Single request	HTTP 201 Created response is received
3	Response time observation	1	Single request	Response completes and time is recorded
4	API/server connectivity check	1	Single request	Local API responds without connection failure

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)
1	Response Time (Avg)	< 2 seconds	10.72 seconds*	Not applicable to load test
2	Response Time (Max)	< 5 seconds	10.72 seconds*	Not applicable to load test
3	Throughput (Req/sec)	—	Not measured	Pending
4	Error Rate	< 1%	0% for shown request*	Not applicable to load test
5	CPU Utilization	< 80%	Not measured	Pending
6	Memory Utilization	< 80%	Not measured	Pending

*The 10.72-second value is the single API request time visible in the supplied screenshot; it is not an average/max from a multi-user load test.

Step 4: Observations & Analysis

The AI blog generation endpoint was tested locally through Thunder Client. The supplied output shows a successful HTTP 201 Created response and generated blog content. The displayed response time is 10.72 seconds for the shown request. A complete load/performance test with multiple virtual users is required to calculate reliable average response time, throughput, error rate, CPU utilization and memory utilization.

Step 5: Screenshots / Evidence

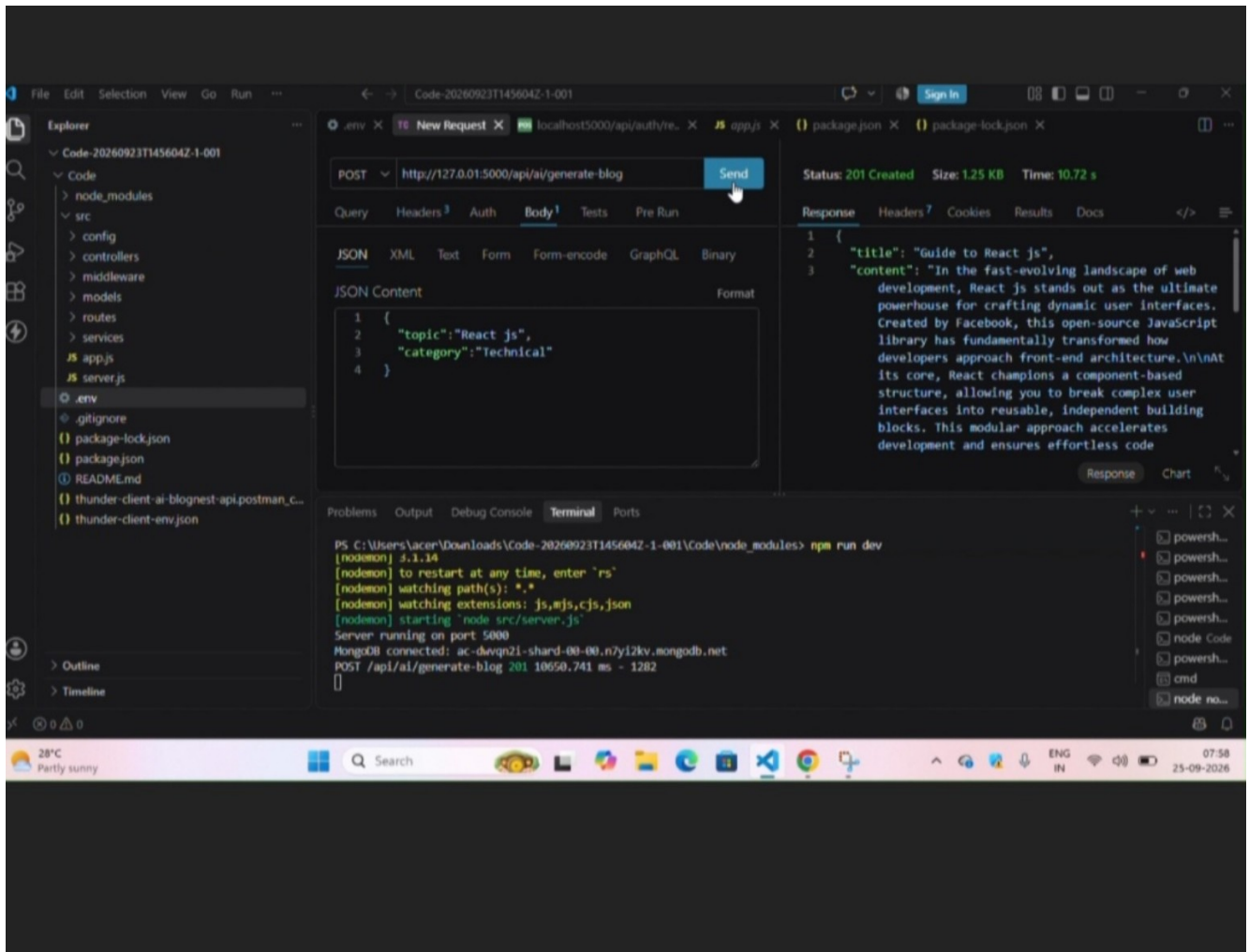


Figure 1: AI Blog Generation API – successful 201 Created response with 10.72-second response time.